

BRASOV-GHIMBAV, RomaniaWMO: **153000**Lat: **45.6958N**Lon: **25.5261E**Elev: **1756**StdP: **13.79**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.2	7.0	-1.4	5.4	2.1	3.8	7.1	8.3	29.2	37.4	22.8	35.5	1.8	230	0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	24.3	86.2	67.0	83.4	66.2	80.7	65.5	69.5	81.7	68.1	79.7	66.8	77.8	7.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.6	100.8	73.9	64.2	95.9	71.9	62.9	91.7	71.1	34.7	82.0	33.4	79.7	32.4	77.8	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	19.5	17.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.2	25.3	32.1	38.7	47.8	57.8	64.8	66.2	66.9	58.4	48.9	39.5	30.6
	DBStd	16.01	9.71	9.38	8.14	8.76	5.59	5.04	4.76	4.71	7.23	7.03	8.65	8.24
	HDD50	2824	765	502	355	143	11	1	0	0	13	105	328	601
	HDD65	6399	1230	922	814	517	227	64	45	36	215	499	764	1066
	CDD50	2158	0	1	6	76	253	445	502	524	264	72	15	0
	CDD65	256	0	0	0	0	5	58	82	95	16	0	0	0
	CDH74	2961	0	0	0	24	141	607	796	1088	296	9	0	0
	CDH80	739	0	0	0	0	12	122	204	333	68	0	0	0
Wind	WSAvg	5.6	4.6	5.6	7.2	7.0	5.7	5.6	5.8	5.0	5.6	5.1	5.3	5.2
Precipitation	PrecAvg	23.4	1.1	0.9	1.1	1.8	2.9	3.5	3.7	2.8	1.8	1.4	1.3	1.1
	PrecMax	31.6	2.7	1.7	3.0	3.9	6.7	6.3	7.2	5.2	4.8	5.4	3.0	2.3
	PrecMin	15.1	0.2	0.0	0.1	0.8	0.6	1.2	0.8	0.6	0.1	0.2	0.2	0.0
	PrecStd	3.9	0.7	0.4	0.8	0.7	1.4	1.3	1.6	1.3	1.2	1.0	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.6	61.1	67.6	76.9	81.7	86.2	88.4	90.4	86.5	74.0	66.4	53.5
		MCWB	44.0	51.8	51.4	57.8	60.8	67.4	67.5	69.3	65.3	58.4	54.0	47.2
	2%	DB	43.5	53.2	61.3	73.4	78.2	83.5	85.1	86.8	81.0	70.7	61.2	48.8
		MCWB	39.4	45.7	48.8	57.0	60.8	67.4	67.5	67.0	63.3	56.7	52.3	44.3
	5%	DB	39.3	49.1	56.3	68.3	74.5	80.6	82.0	83.5	76.7	66.4	56.7	44.7
		MCWB	36.6	43.3	46.2	54.3	59.6	66.3	66.7	65.9	61.7	55.2	49.8	41.1
	10%	DB	36.7	44.7	51.4	63.7	71.1	77.5	78.5	80.4	72.6	61.6	52.9	40.6
		MCWB	34.8	40.6	43.3	52.1	58.4	65.3	65.8	65.5	60.4	53.4	47.6	38.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.2	51.5	52.8	59.6	64.9	71.7	70.9	71.1	67.1	60.6	54.8	48.0
		MCDB	49.3	60.0	66.4	73.0	76.0	82.9	82.2	85.4	81.9	69.9	63.7	51.4
	2%	WB	39.9	46.5	49.3	57.6	62.5	69.0	69.2	69.4	64.8	58.1	52.9	45.2
		MCDB	42.9	52.5	60.4	70.7	73.5	79.8	80.6	82.8	76.5	67.5	60.6	48.9
	5%	WB	37.2	43.9	46.9	55.3	60.7	67.4	67.9	67.7	62.8	56.3	50.9	41.6
		MCDB	39.3	48.7	55.7	67.4	71.6	78.2	78.9	79.9	74.6	64.4	56.9	44.8
	10%	WB	35.0	40.7	44.5	53.1	59.2	66.0	66.6	66.2	61.4	54.2	48.1	38.7
		MCDB	36.5	44.3	50.4	62.2	68.5	76.2	76.4	77.8	71.5	60.9	52.5	40.8
Mean Daily Temperature Range	5% DB	MDBR	13.6	16.4	17.7	20.9	21.0	21.4	21.4	24.3	21.3	20.6	15.7	11.2
		MCDBR	14.1	22.1	27.4	30.0	28.7	26.3	28.1	30.6	30.2	30.8	22.8	16.7
		MCWBR	10.3	14.8	15.8	15.1	13.5	12.1	12.4	12.5	13.4	17.4	14.2	12.5
	5% WB	MCDBR	13.4	22.0	25.3	27.3	25.1	24.7	24.7	27.1	27.2	27.8	21.8	16.2
		MCWBR	10.1	14.8	14.9	14.1	12.4	11.8	11.3	11.9	12.5	16.1	13.5	12.3
Clear-Sky Solar Irradiance	taub	0.299	0.327	0.369	0.426	0.408	0.408	0.424	0.436	0.392	0.355	0.329	0.300	
	taud	2.341	2.276	2.242	2.136	2.255	2.289	2.248	2.211	2.307	2.402	2.423	2.384	
	Ebn at Noon	254	267	271	264	273	273	266	258	260	255	241	240	
	Edh at Noon	24	31	38	46	42	41	42	42	35	28	23	21	
All-Sky Solar Radiation	RadAvg	437	699	1039	1360	1661	1797	1819	1652	1205	833	484	352	
	RadStd	47	86	111	147	161	162	157	141	121	105	51	42	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (74 neighbors)	+1.08	N/A	N/A	+1.24	N/A	N/A	-238	-320	+137	+45	

Nomenclature: See separate page